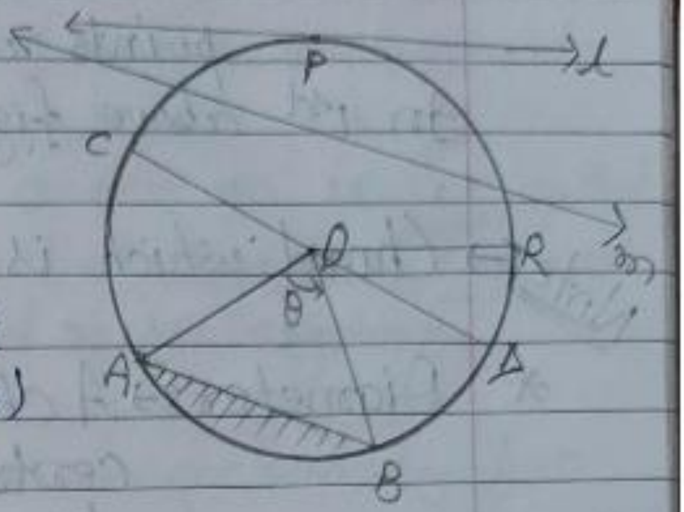
Circle

O ⇒ centre
OR ⇒ Radius
AB ⇒ chord
$\widehat{AB}$ ⇒ Arc
ABO ⇒ Sector
Area inclosed AB ⇒ Segment
$\angle AOB$ ⇒ Central angle (θ)
CD ⇒ Diameter
Line l ⇒ Tangent
~~Line~~ point P ⇒ point of contact
Line m ⇒ Secant.



* Circle :→ A set of all points on a plane which are equidistant from a fixed given point.

→ Here, fixed given point (O) is called centre of a circle.

* Radius :→ Line segment joining from centre to any point on a circle. is called radius.

In 1st above figure OR is radius.

→ It is denoted by small 'r'.

Note :→ A circle whose centre 'O' and radius 'r' is denoted by 'c(O, r)'.

*. Chord:→ Line segment joining two distinct points on a circle is called chord.
In 1st above figure $\overline{AB}$ is chord.

Note:→ Chord which is nearer to centre is larger.

* Diameter:→ A chord which passes through centre of a circle is called diameter.

→ Diameter is the longest chord of a circle.

→ Diameter is twice of radius.

* Arc:→ A continuous part of a circle is called arc.

→ An arc whose length is less than the semi circle is called minor arc.

→ An arc whose length is more than the semi circle is called major arc.

→ It is denoted by '$\overset{\frown}{AB}$'

* Segment:→ Part of circle enclosed by a chord and corresponding arc is called segment.

*→ Minor and major segment:→ A segment enclosed by a chord and minor arc is called minor segment

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A segment enclosed by a chord and a major arc is called major ~~arc~~ segment.

Or

A segment enclosed centre in it is called major segment and A segment ~~does~~ dose not enclosed centre in it is called minor segment.

* Sector :- A part of circle enclosed by two radii and corresponding arc is called sector.

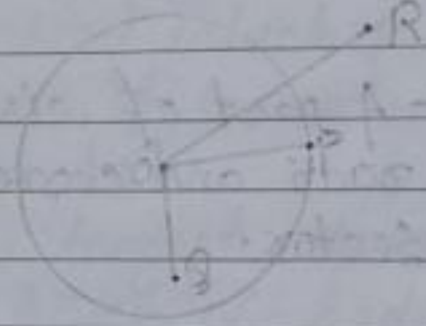
* Minor and major sector :- A sector whose central angle is less than 180° is called minor sector.
And A sector whose central angle is more than ~~one~~ 180° is called major sector.

Note :- Circles divides the plane in three parts.
(i) Interior of a circle.
(ii) Exterior of a circle.
(iii) Boundary of a circle.

* (i) Interior of a circle :- The region which consists all point lie inside of the circle.

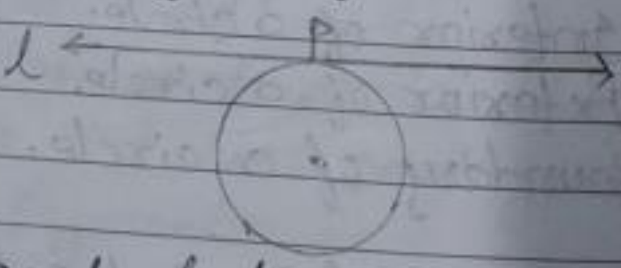
*→ (ii) Exterior of circle:- The region which consists all the points lies outside of the circle.

*→ (iii) Boundary of a circle:- The region which consists all the points lies on the circle.



→ (i) $OQ < r$, then Q must lie interior of a circle.
→ (ii) $OP = r$, then P must lie on the circle.
→ (iii) $OR > r$, then R must lie exterior of a circle.

* Tangent of a circle:- A line which intersect the circle at a point is called tangent of a circle.

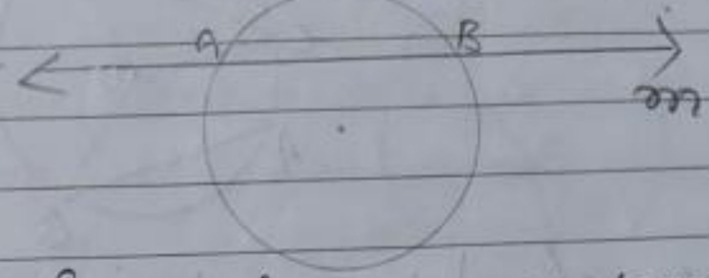


→ line l is tangent of the circle.

Note:- → Common point of circle and tangent is called "point of contact".
→ Here, P is point of contact.

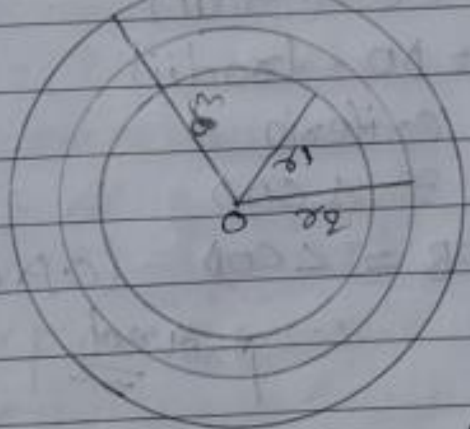
* Secant of a circle :- A line which intersect the circle at two distinct point is called secant of a circle.

→ line m intersect at two distinct point A and B.



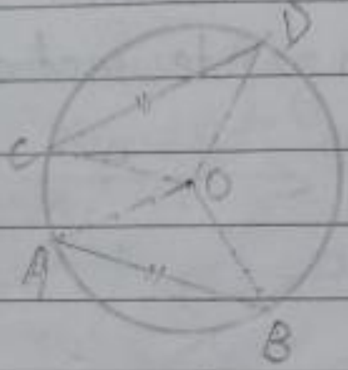
For given figure, line m is secant of the circle.

* Concentric circle :- Two or more circles having a common centre are called concentric circle.



Here, $c(0, r_1)$, $c(0, r_2)$ and $c(0, r_3)$ are concentric circles.

Theorem 1st :→ Equal chords of a circle subtend equal angle at centre.



Given :→ $c(O, r)$ is a circle in which chord $AB = CD$

To prove :→ $\angle AOB = \angle COD$

Proof :→ In ΔAOB and ΔCOD

$AB = CD$ (given)

$AO = CO$ (radii)

$BO = DO$ (radii)

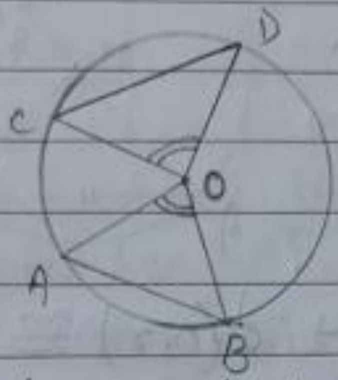
By S.S.S. criteria

$\Delta AOB \cong \Delta COD$

$\therefore \angle AOB = \angle COD$ (C.P.C.T)

proved

☞ Theorem 2nd :→ (converse of previous theorem 1st)
:→ If two chord of a circle subtend equal angles at centre, then the chords are equal.



Given :→ $c(O,r)$ is a circle in which AB and CD are two chords such that $\angle AOB = \angle COD$

To prove :→ $AB = CD$

proof :→ for ΔAOB and ΔCOD

$AO = CO$ (radii)

$\angle AOB = \angle COD$ (given)

$BO = DO$ (radii)

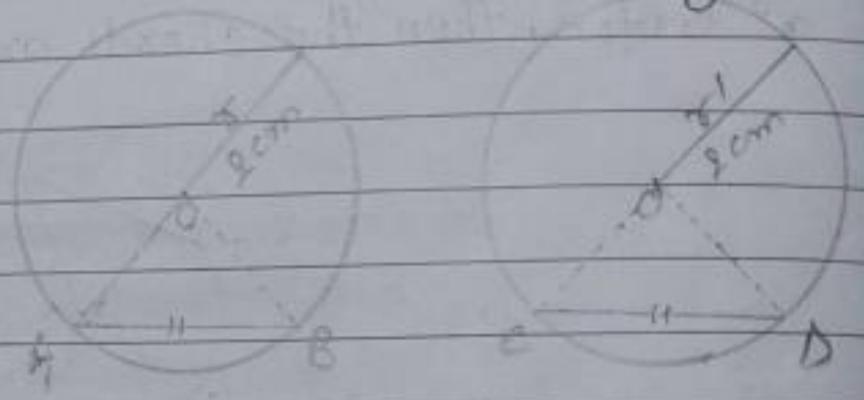
By S.A.S. criteria.

$\Delta AOB \cong \Delta COD$

$\therefore AB = CD$ (C.P.C.T)

proved

Note:→ Theorem:→ Equal chords of two congruent circles subtend equal angles at centre.



Given:→ $c(O, r) \cong c(O', r')$ in which chord $AB = CD$.

To prove:→ $\angle AOB = \angle COD$

proof:→ $AO = OC$ (radius of two congruent circle)

$OB = OD$ (,,)

$AB = CD$ (given)

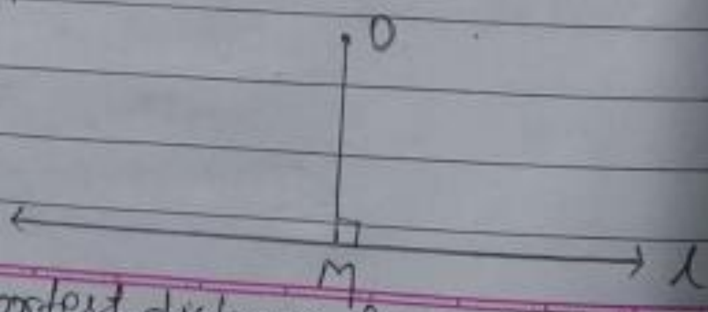
By S.S.S. criteria

$\Delta AOB \cong \Delta COD$.

$\therefore \angle AOB = \angle COD$ (C.P.C.T)

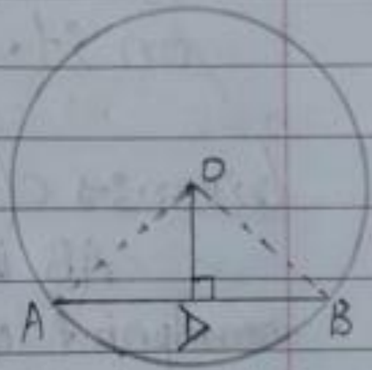
proved

Note:→ From a given point to a given line, perpendicular distance is shortest distance between point and line.



→ OM is shortest distance from point 'O' to line 'l'

Theorem 3rd :→ Perpendicular from centre of a circle to the chord, bisect the. Chord.



Given :→ $c(O, r)$ is a circle in which $OD \perp AB$.

To prove :→ $AD = BD$

Construction :- Join AO and BO

proof :→ In right angle triangle AOD and BOD

$AO = BO$ (radii) (hypotenuse)

$OD = OD$ (common)

By. R.H.S criteria.

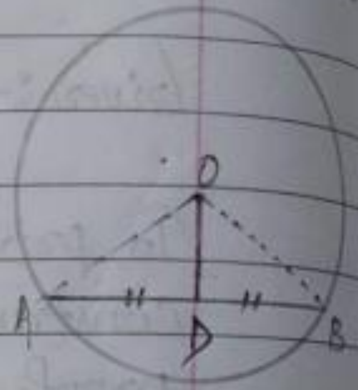
$\triangle AOD \cong \triangle BOD$

$\therefore AD = BD$ (C.P.C.T)

proved

Theorem 4th :→ (converse of previous theorem 3rd)

:→ Line segment joining from centre of a circle to midpoint of a chord is perpendicular on it.



Given :→ $C(O, r)$ is a circle in which AB is chord and D is midpoint of chord AB.

To prove :→ $OD \perp AB$.

Construction :→ Join AO and BO.

Proof :→ In ΔAOD and ΔBOD

$OA = OB$ (radii)

$OD = OD$ (common)

$AD = BD$ (given)

By S.S.S criteria.

$\Delta AOD \cong \Delta BOD$

$\therefore \angle ADO = \angle BDO$ (C.P.C.T) eqn – (1)

By linear pair

$\angle ADO + \angle BDO = 180^\circ$

$\angle ADO + \angle ADO = 180^\circ$ { from eqn – (1) }

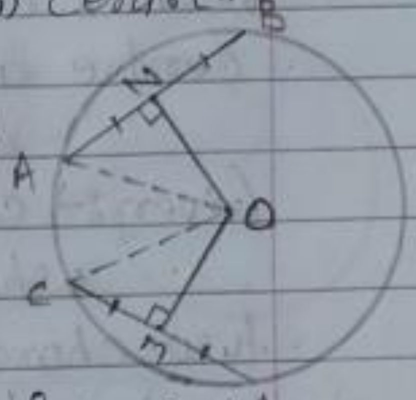
$2\angle ADO = 180^\circ$

$\angle ADO = 90^\circ$

$\therefore OD \perp AB$

proved

Theorem 5th :→ Equal chord of a circle are equidistant from centre.



Given :→ c(O,r) is a circle in which AB and CD are two equal chords (AB = CD)

To prove :→ Chords are equidistant from centre.
or, OM = ON.

Construction :→ Join AO and CO.

proof :→ AB = CD (given)

$\frac{1}{2}AB = \frac{1}{2}CD$ (multiplying $\frac{1}{2}$ both side)

AN = CM {perpendicular from centre to chord, bisect the chord}

In ΔCOM and ΔAON

AN = CM

AO = CO (radii) (hypotenuse)

By R.H.S. criteria.

ΔCOM ≅ ΔAON

∴ OM = ON (C.P.C.T)

proved

Theorem 6th :→ (converse of previous theorem 5th

:→ Two chords which are equidistant from centre are equal to each other.

Given :→ c(O, r) is a circle in which AB and CD are two chords equal distance from centre. (ON = OM)

To prove :→ AB = CD

Construction :→ Join AO and CO

proof :→ In Δ COM and Δ AON

ON = OM (given)

AO = CO (radii) (hypotenuse)

∠CMO = ∠ANO (90°)

By R.H.S criteria.

Δ COM ≅ Δ AON.

∴ CM = AN.

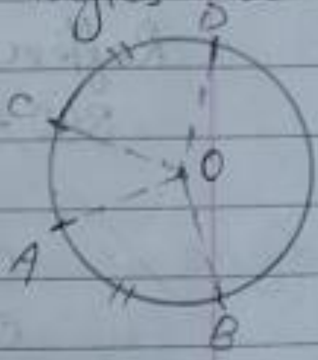
2CM = 2AN (multiplying 2 both side)

~~AB = CD~~

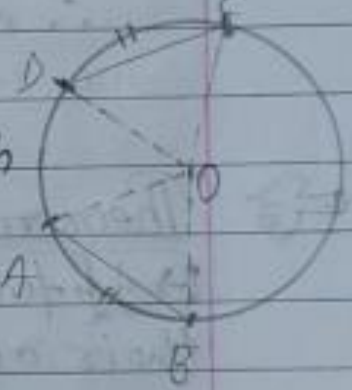
CD = AB. proved

Note:- If two arcs of a circle and are congruent then their corresponding central angles are equal.

If $\widehat{AB} \cong \widehat{CD}$, then $\angle AOB = \angle COD$.



Theorem 7th:- If the two arcs of the circle are congruents then their corresponding chords are equal.



Given:- $C(O,r)$ is a circle in which $\widehat{AB}$ and $\widehat{CD}$ are two arc such that $\widehat{AB} \cong \widehat{CD}$

To prove:- $AB = CD$

Construction:- OA, OB, OC and OD

Proof:- Case 1st:- If $\widehat{AB}$ and $\widehat{CD}$ are minor arcs.

$\widehat{AB} \cong \widehat{CD}$ (given)

$\therefore \angle AOB = \angle COD$

$OB = OD$ (radius)

$OA = OC$ (radius)

By S.A.S criteria.

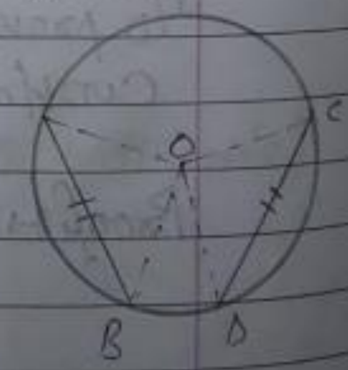
$\Delta AOB \cong \Delta COD$

$\therefore AB = CD$ (C.P.C.T) proved.

Case II^{nd} :→ when $\overparen{AB}$ and $\overparen{CD}$ are major arc, $\overparen{AB} \cong \overparen{CD}$

$$\text{Reflex } \angle AOB = \text{Reflex } \angle COD$$
$$\cancel{360} + \angle AOB = \cancel{360°} + \angle COD$$
$$\angle AOB = \angle COD.$$
$$OA = OC$$
$$OB = OD$$
By S.A.S criteria.
$$\therefore \triangle AOB \cong \triangle COD.$$
$$\therefore AB = CD \quad (C.P.C.T)$$
proved

Theorem 8^{th} :→ (Converse of previous theorem 7^{th})
:→ If two chords of a circle are equal then their corresponding arcs (minor, major or semicircular) are congruent



Given :→ $C(O, r)$ is a circle in which AB and CD are two chords.
Such that $AB = CD$.
To prove :→ $\overparen{AB} \cong \overparen{CD}$
Construction :→ Join OA, OB, OC and OD.

Proof :→ Case I^{st} :→ For minor arcs.
In $\triangle AOB$ and $\triangle COD$
$AB = CD$ given.
$\left.\begin{array}{l} OA = OC \\ OB = OD \end{array}\right\}$ radius

By S.S.S. criteria.

$\triangle AOB \cong \triangle COD$.

$\therefore \angle AOB = \angle COD$

$\therefore \overset{\frown}{AB} \cong \overset{\frown}{CD}$

proved

Case IInd :- when $\overset{\frown}{AB}$ and $\overset{\frown}{CD}$ are major arcs.

In $\triangle AOB$ and $\triangle COD$.

$AB = CD$ (given)

$OA = OC$ } radius

$OB = OD$

By S.S.S. criteria.

$\triangle AOB \cong \triangle COD$.

$\therefore \angle AOB = \angle COD$.

$360 - \text{Ref.}\angle AOB = 360 - \text{Ref.}\angle COD$

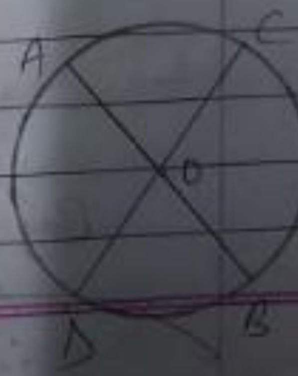
$\text{Ref}\angle AOB = \text{Ref}\angle COD$

$\therefore$ Major $\overset{\frown}{AB} \cong$ Major $\overset{\frown}{CD}$

proved

Case IIIrd :- For Arcs of Semi circles.

If $\overset{\frown}{AB}$ and $\overset{\frown}{CD}$ are arcs of semi-circle then AB and CD are Diameter.



Given :- $AB = CD$

$\frac{1}{2}AB = \frac{1}{2}CD$.

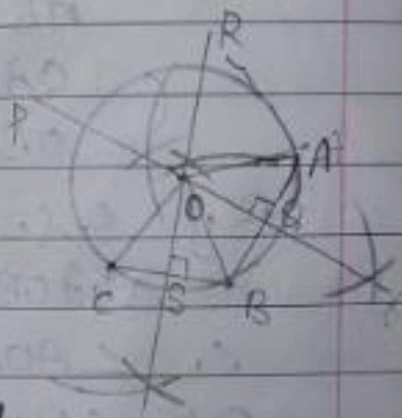
$OA = OC$.

radius are equal then semi circles with radius OA and OC are congruent.

$\overset{\frown}{AB} \cong \overset{\frown}{CB}$

Proved

√·V·I Theorem 9th :→ There is one and only one circle passes through three non collinear point.



Given :→ A, B, and C are three non-collinear points.

To prove :→ One and only one circle passes through A, B, and C.

Construction :→ Join AB and BC

proof :→ Draw PQ and RS, Perpendicular bisector of AB and BC, respectively which intersect at O. Join OA, OB and OC.

In ΔOCS and ΔOBS.

$CS = BS$ (bisector)

$\angle OSC = \angle OSB$ (90°)

$OS = OS$ (common)

By S.A.S criteria.

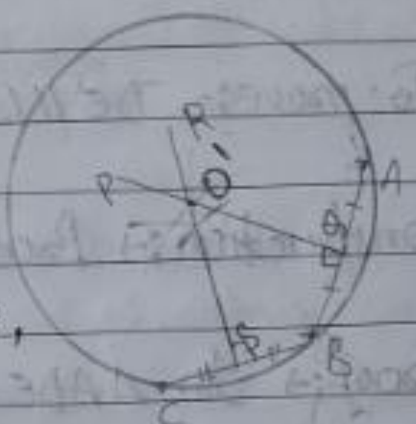
$\Delta OCS \cong \Delta OBS$.

$\therefore OC = OB$ (C.P.C.T) —— (1)

Similarly
$OB = OA$ —— (ii)
From (i) and (ii)
$OA = OB = OC$
~~also~~ With O as centre and $OA = OB = OC$ as radius. Draw a circle $c(O, r)$ which passes through, A, B, and C

Proved

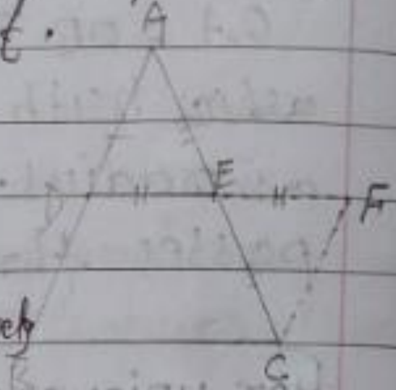
For uniqueness



If possible, let, another circle $c(O', r')$ also passes through point A, B, and C.
$\therefore$ AB and BC are chords. of a circle $c(O' r')$.
PQ and RS are perpendicular bisector of AB and BC.
then point of intersection of PQ and RS is centre O'.
But given PQ and RS intersect at O.
we know that two lines intersect at one and only one point.
$\therefore$ O' coincide on O.
$\therefore$ $O'A = OA$
$r' = r$
$\therefore$ $c(O' r')$ coincide on $c(O, r)$
$\therefore$ Both circle are same. proved

V.V.I ** B.M

Midpoint Theorem of Triangle :- Line segment joining midpoint of two sides of a triangles is parallel to third side and half of it.



Given :- ABC is a triangle in which D and E is midpoint of AB and AC respectively

To prove :- $DE \parallel BC$ and $BC = 2DE$

Construction :- Produce DE to F such that $DE = EF$. Join FC.

Proof :- In $\triangle ADE$ and $\triangle EFC$.

$DE = EF$ (Construction)

$AE = EC$ (given)

$\angle AED = \angle FEC$ (V.O.A)

By S.A.S criteria.

$\triangle ADE \cong \triangle EFC$

$\therefore \angle F = \angle ADE$ } (C.P.C.T)

and $FC = AD$ }

$\angle F$ and $\angle ADE$ are pair of Alt. interior angle.

$\therefore DB \parallel FC$ — (i)

$FC = AD$ (from c.p.c.t)

$AD = DB$ (given)

$\therefore DB = FC$ — (ii)

From eqn (i) and (ii)
If one pair of opposite sides are parallel and equal then quadrilateral is ||gm.

$\therefore$ DFCB is ||gm.

DF || BC

$\therefore$ DE || BC

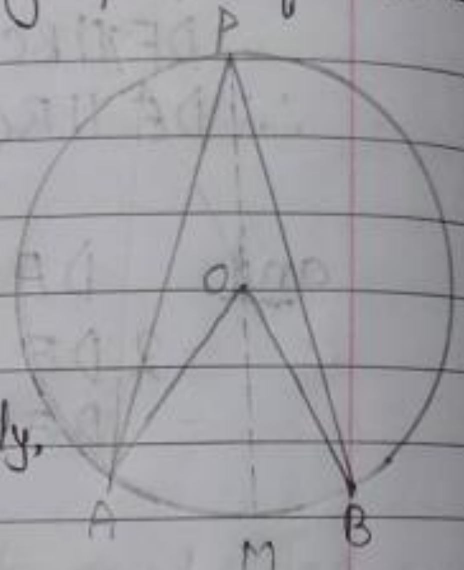
proved

and $DF = BC$

$2DE = BC$.

$DE = \frac{1}{2} BC$. proved

Theorem 10th :→ The angles subtended by an arc of a circle at the centre is double then the angles subtended by it at any point on the remaining part of circle.



Given :→ $c(0, r)$ is a circle in which $\widehat{AB}$ is an arc $\angle AOB$ and $\angle APB$ are angles on the centre and on the circle poit p . respectively.

To prove :→ $\angle AOB = 2\angle APB$

Construction :→ Draw a line PM which passes through O.

proof :→ In $\Delta\, APO$

$OP = OA$

$\angle OPA = \angle OAP = x$ (let)

using exterior angle property of triangle.

$\angle AOM = \angle OPA + \angle OAP = x + x = 2x$

In $\Delta\, OPB$

$OP = OB$

$\angle OPB = \angle OBP = y$ (let)

using exterior angle property of triangle

$\angle BOM = \angle OPB + \angle OBP = y + y = 2y$

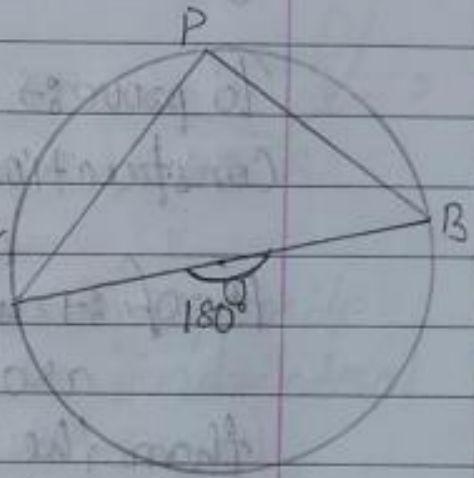
$\angle APB = x + y$ —— ①

$\angle AOB = \angle AOM + \angle BOM$

$\angle AOB = 2x + 2y$
$\angle AOB = 2(x+y)$
$\angle AOB = 2(\angle APO + \angle BPO)$
$\angle AOB = 2\angle APB$

proved

Theorem 11th :- Angle on semicircle is right angle.



Given :- $c(o,r)$ is a circle in which AB is a diameter
$\angle APB$ is an angle on semicircle.

To prove :- $\angle APB = 90°$

proof :- we know that diameter AB is a straight line.

$\therefore \angle AOB = 180°$

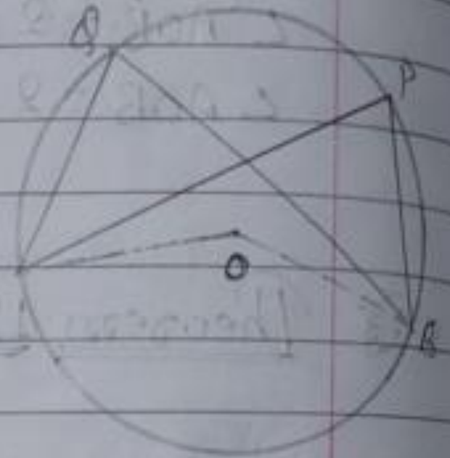
$\angle APB = \frac{1}{2}\angle AOB$ {Angle subtand at centre by an arc is double than the angle subtand by it on circle}

$\angle APB = \frac{1}{2} \times 180° = 90°$

$\angle APB = 90°$

proved

Theorem 12th :→ Angles in the same segment of a circle are equal.



Given :→ $c(o, r)$ is a circle $\angle APB$ and $\angle AQB$ are angles on same segment of the circle.

To prove :→ $\angle APB = \angle AQB$.

Construction :→ Join AO and BO

Proof :→ we know that angle subtended by an arc of a circle at centre is double than the angle subtended by it at any point on the circle.

$$\angle APB = \frac{1}{2} \times \angle AOB \quad \text{— (1)}$$

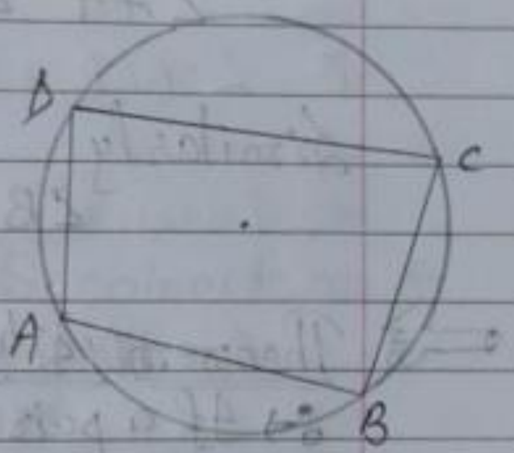
$$\angle AQB = \frac{1}{2} \times \angle AOB \quad \text{— (11)}$$

R.H.S of both eqn is equal the L.H.S is must be equal.

$\therefore \angle APB = \angle AQB$ [from eqn (1) and (11)]

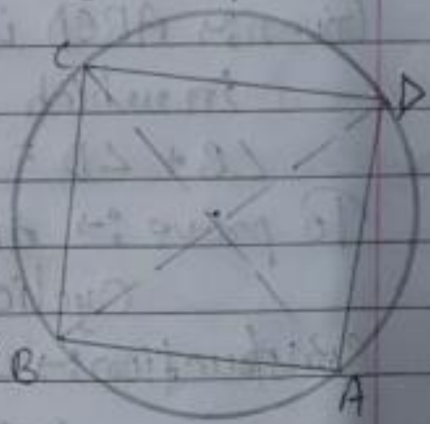
proved

** Cyclic quardrilateral :→ A quardrilateral whose all four vertices lies on the circle is called cyclic quardrilateral.



Theorem 13^{th} :→ The sum of pair of opposite angles of a cyclic quadrilateral is 180°.

Given :→ ABCD is a cyclic quadrilateral.



To prove :→ (i) $\angle A + \angle C = 180^\circ$

(ii) $\angle B + \angle D = 180^\circ$

Construction :→ Join AC and BD

Proof :→ we know that angles on same segment are equal.

$\angle BAC = \angle BDC$ — (1)

$\angle CAD = \angle CBD$ — (11)

for ΔBCD

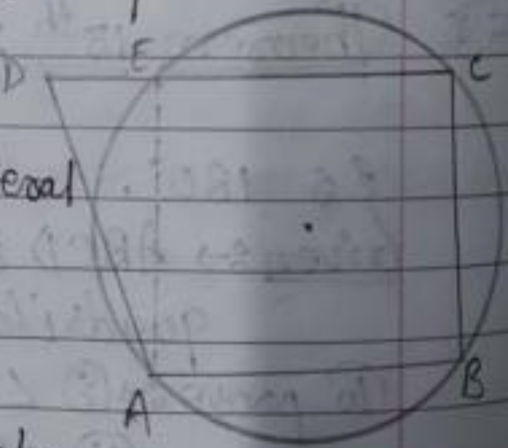
$\angle BDC + \angle CBD + \angle BCD = 180^\circ$

Putt in the value of eqn - (i) and (ii)

$$\angle BAC + \angle CAD + \angle BCD = 180^\circ$$
$$\angle A + \angle C = 180^\circ$$
proved

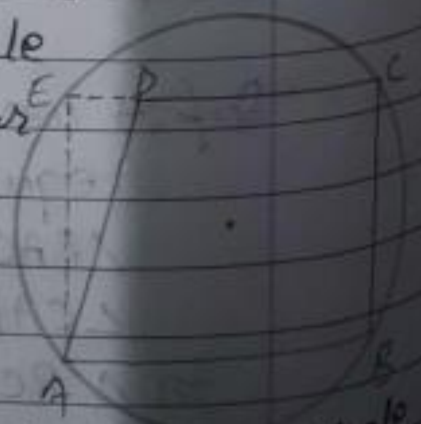
Similarly
$$\angle B + \angle D = 180^\circ$$ proved

☞ <u>Theorem 14th</u> :→ (converse of previous theorem 13th)
:→ If a pair of opposite angles of quadrilateral is supplimentry then the quadrilateral is cyclic.



Given:→ ABCD is a quadrilateral in which
$\angle B + \angle D = 180^\circ$ —— (1)
To prove :→ ABCD is a cyclic quadrilateral.
Construction:→ we know that there is one and only one circle passes through three non-collinear point.
Draw a circle c(o,r) passes through A, B and C.
If possible, let D not lies on c(o,r) and CD or produced CD intersect the circle

<u>or</u>



... join AE.

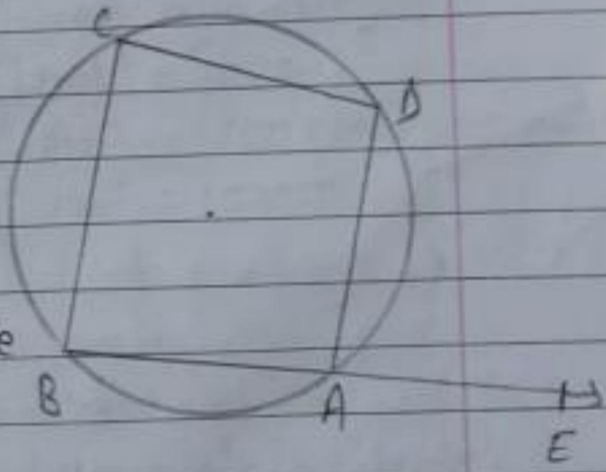
Proof:→ ABCE is a cyclic quadrilateral. ~~opposite~~
Opposite angles of cyclic quadrilateral is supplementary.
$\therefore \angle B + \angle E = 180°$ ——— (ii)
from (i) and (ii)

$\angle B + \angle D = \angle B + \angle E$

$\angle D = \angle D$

it is only possible when D coincide on E.
$\therefore$ ABCD is a cyclic quadrilateral.
proved

Theorem 15th :→ Exterior angle property of cyclic quadrilateral:→ If any one side of a cyclic quadrilateral be produce then exterior angle so formed is equal to interior opposite angle.



Given:→ ABCD is a cyclic quadrilateral and side BA produced to E such that exterior angle $\angle DAE$ formed.

To prove:- $\angle BCD = \angle DAE$.

Proof:→ opposite angle of cyclic quadrilateral is supplementary.

$\therefore \angle BAD + \angle BCD = 180° —— (1)$

$\angle BAD + \angle DAE = 180° —— (ii)$ {linear pair}

from (i) and (ii)

$\cancel{\angle BAD} + \angle BCD = \cancel{\angle BAD} + \angle DAE$

$\angle BCD = \angle DAE$

proved